

Frontend Data Visualization: Dashboard Report

(Dynamic Views)

Project Report: Supply Chain Resilience & Delivery Intelligence Dashboard

(Proactive Attrition Logic & Predictive Modeling)

Industry Partner: APL Logistics

Organization: Unified Mentor Pvt. Ltd. (Internship 2026)

Tool Used: Tableau Public Visuals & Tableau Formula

Theme: Professional Analytics Theme (As per Dashboard Visuals – Optimized for Clarity and Data-Driven Insights)

Executive Summary

The APL Logistics Delivery Risk Prediction & Geographical Scope Dashboard acts as the primary strategic instrument for maintaining supply chain reliability. By integrating multi-dimensional data—ranging from shipping mode efficiency to regional delivery bottlenecks—this report provides a comprehensive view of logistics health. **A key innovation in this suite is the integrated "Dynamic Alerting System," which transforms the dashboard from a passive reporting tool into an active, real-time operational monitor.**

1. Dashboard 1: Global Logistics Risk Map & KPI Overview (45% Threshold)

- **Focus:** Global hotspots of late deliveries.
- **Key Logic:** 'Late Risk %' >45% flagged as High Risk. The 45% threshold creates an "early warning buffer," allowing proactive intervention before failure becomes certain.

2. Dashboard 2: Delivery Gap & Risk Trends Overview

- **Focus:** Mapping 'Avg. Delivery Gap (Hours)' against shipping modes (Same Day, First Class, etc.) to pinpoint efficiency leaks.

3. Dashboard 3: Geographical Scope & Regional Risk Distribution Insights

- **Focus:** Visualizing intensity of delivery gaps via Regional Heat Maps and identifying high-risk regions requiring resource intervention.

4. Dashboard 4: Predictive SLA Governance & Dynamic Risk Mitigation

- **Focus:** End-to-end breach monitoring and regional criticality scoring.
- **New Feature: Intelligent Operational Alert System:**
 - To bridge the gap between "Insight" and "Action," this dashboard features a **Dynamic Alerting System** using **Dynamic Zone Visibility**.

- **Logic:** As soon as an order or region breaches the >45% High-Risk or >65% Critical-Risk threshold, a prominent **Red Alert Banner** dynamically appears at the top of the dashboard.
- **Operational Impact:** This "Smart UI" eliminates manual monitoring. Warehouse managers receive instant visual feedback, shifting the operational strategy from reactive "firefighting" to immediate, data-driven prevention. Once the risk is mitigated, the alert automatically vanishes, ensuring a clean, distraction-free interface.
- **Risk Segmentation:** Critical (>65%), High (>45%), and Safe (<45%).

Future Scope

- **AI/ML Integration:** Incorporating Random Forest or XGBoost models to predict delays based on real-time weather/traffic.
- **Automated Alerting System:** Scaling the current dashboard-based alert to real-time email or SMS notifications for warehouse managers.
- **Vendor Performance Tracking:** Expanding analytics to specific transport vendors for better negotiation.
- **Cost-Impact Analysis:** Linking delivery delays directly to financial penalties to prioritize ROI-focused interventions.

Conclusion

The integration of the Dynamic Alerting System within the SLA Governance Dashboard ensures that leadership can identify and resolve logistics bottlenecks with absolute precision. By proactively addressing regions that fall outside the "Safe" threshold, APL Logistics can minimize SLA penalties, optimize delivery performance, and significantly bolster customer trust.